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ITAI 4370: AI 5/6G Comm & ORAN Net

Assignment 02

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Question 1 - *Explain the difference between EPC (4G) and 5GC (5G).*

EPC is the 4G core network, built as a fixed set of dedicated boxes, each with a set job such as mobility, session handling, or data routing, all wired together in a rigid way. 5GC is the 5G core, built instead as small software services that run in the cloud and call each other only as needed, which makes it much easier to scale, update, and tailor for uses like network slicing (RCR Wireless News, 2021).

Examples:

- In EPC, mobility and session jobs live in dedicated nodes like the MME and the gateways, which are hard to change once they are deployed.
- In 5GC, those same jobs become independent services such as the AMF and the SMF, which can be scaled or updated on their own.

Question 2 - *Why is network slicing important in 5G? Provide real-world examples.*

Network slicing lets one operator run several separate virtual networks on top of a single shared physical network, with each slice tuned for a particular need. It is important because 5G must serve very different demands at the same time, from high-speed video to crowds of small sensors to split-second control, and one general-purpose network cannot meet all of them well (NETSCOUT, n.d.).

Examples:

- A factory can run a low-delay slice for real-time robot control and a separate slice for its many low-power sensors.
- A hospital can use a secure, isolated slice for remote surgery so the connection stays reliable and the delay stays very low.

Question 3 - *What is Multi-access Edge Computing (MEC), and how does it reduce latency?*

Multi-access Edge Computing (MEC) is an ETSI design that puts cloud-style computing power at the edge of the network, close to the user, such as at a cell tower instead of in a distant central data center. It lowers latency because the data travels a much shorter distance to be processed and answered, so the round trip is faster and the core network carries less traffic (ETSI, n.d.).

Examples:

- A self-driving car can send sensor data to a nearby edge server and get a reply quickly enough for split-second decisions.

- An augmented reality headset can process its heavy video at the edge so the images keep up with the user's movements.

Question 4 - *Describe the role of AI in dynamic resource allocation.*

Dynamic resource allocation is the network deciding, moment to moment, how to share out limited resources such as spectrum, bandwidth, and computing power as demand changes. AI helps by learning from past and live network data to predict what is coming and adjust those resources in real time, which holds service quality steady and uses the network more efficiently than fixed manual rules (IEEE 5G/6G Innovation Testbed, n.d.).

Examples:

- A model can forecast a busy period and move extra bandwidth to a slice before the traffic actually rises.
- AI can detect quiet hours and place some base stations into sleep mode to save energy without harming service.

Question 5 - *In SBA, what is the role of the Network Repository Function (NRF)?*

In the 5G Service-Based Architecture, the network functions are software services that need a reliable way to find each other before they can work together. The Network Repository Function (NRF) is the central directory that makes this possible, since each function registers itself and the services it offers, and any function that needs a service looks it up through the NRF (Microsoft, 2023).

Examples:

- When a new Session Management Function (SMF) starts up, it registers with the NRF so the rest of the core knows it exists and what it does.
- When an Access and Mobility Management Function (AMF) needs an SMF, it asks the NRF to find an available one and then connects to it.

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